

Rows And Columns

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

You are given a matrix consisting of n rows and m columns.

For each row, compute the sum of its elements. Let **maxRow** be the maximum row sum.

For each column, compute the sum of its elements. Let **maxCol** be the maximum column sum.

Your task is to print **maxRow** and **maxCol**.

Input

The first line contains two integers n and m ($1 \leq n, m \leq 100$).

The next n lines each contain m integers $a_{i,j}$ ($-10^3 \leq a_{i,j} \leq 10^3$).

Output

Print two integers separated by a space:

maxRow maxCol

Examples

standard input	standard output
2 3 1 2 3 4 5 6	15 9
1 1 1	1 1